COMPSCI 689
Lecture 14: Probabilistic Regression

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Data Generating Distributions

- Recall that our assumption about data sets $\mathcal{D}$ is that data cases are generated independently and identically (IID) from an underlying joint probability distribution over $(\mathbf{x}, y)$ pairs:

$$p_*(\mathcal{D}) = \prod_{n=1}^{N} p_*(\mathbf{X}_n = \mathbf{x}_n, Y_n = y_n)$$

- Using the chain rule of probability, we can write:

$$p_*(\mathbf{X}_n = \mathbf{x}_n, Y_n = y_n) = p_*(\mathbf{X}_n = \mathbf{x}_n)p_*(Y_n = y_n|\mathbf{X}_n = \mathbf{x}_n)$$

- We refer to p_* as the *true data generating distribution*. Our only information about p_* is provided through the $(\mathbf{x}_n, y_n)$ *samples* that comprise the data set $\mathcal{D}$.

ERM vs Probabilistic Modeling

- The goal in ERM is to select the parameters θ of a prediction function model that result in the lowest empirical risk.
- As we have seen, minimizing empirical risk approximates minimizing the expected prediction loss under the true data generating distribution p_*.
- An alternative approach to supervised learning is to instead use the data to learn a probabilistic model $p_\theta(Y = y|\mathbf{X} = \mathbf{x})$ that approximates $p_*(Y = y|\mathbf{X} = \mathbf{x})$.
- We can then use the learned model $p_\theta(Y = y|\mathbf{X} = \mathbf{x})$ make a variety of predictions such as the most probable value of y given $\mathbf{x}$ or the expected value of y given $\mathbf{x}$.
- Specifying and learning such models is the focus of probabilistic supervised learning.

Continuous Random Variables

A random variable Z taking values $z \in \mathcal{Z}$ is continuous if it's probability distribution is defined by a probability density function $p : \mathcal{Z} \to \mathbb{R}$ that satisfies the requirements below:

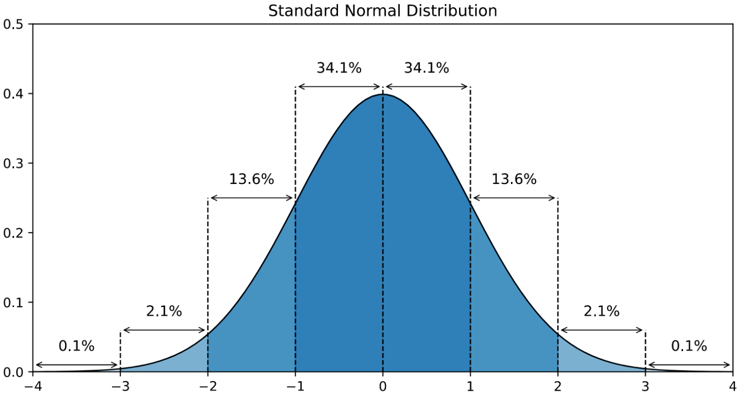
- Normalization: $\int_{\mathcal{Z}} p(Z = z) dz = 1$
- Non-Negativity: $\forall z \in \mathcal{Z}\ p(Z = z) \geq 0$

Example: Standard Normal Random Variable

Consider the standard normal random variable Z.

- Values: $\mathcal{Z} = \mathbb{R}$
- Density Function: $p(Z = z) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2)$

Example: Standard Normal Random Variable



Parametric Probability Density Functions

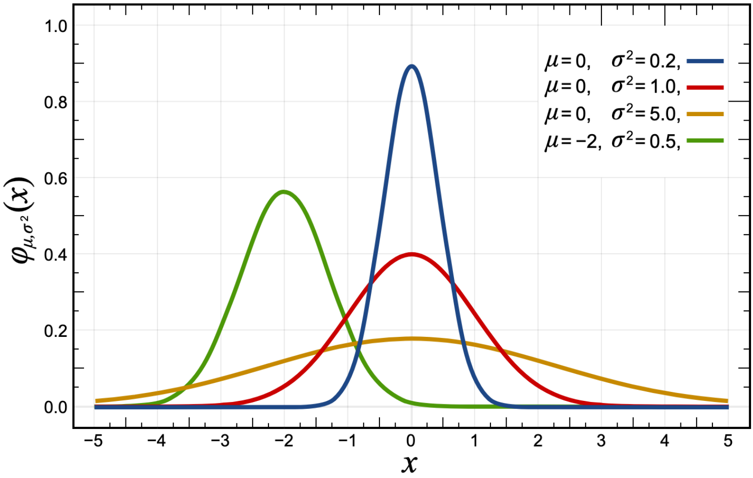
- A parametric probability density function uses a set of parameters to provide access to a family of related probability densities.
- We will denote a parametric probability density function by $p_\phi(Z = z)$ with parameters ϕ.
- We will denote by Φ the parameter space.
- To be valid, a parametric density function must satisfy normalization and non-negativity for all $\phi \in \Phi$

Example: General Normal Random Variable

Consider the general normal random variable Z.

- Values: $\mathcal{Z} = \mathbb{R}$
- Parameters: $\phi = [\mu, \sigma]$, $\mu \in \mathbb{R}$, $\sigma \in \mathbb{R}^{\geq 0}$
- Density Function: $p_\phi(Z = z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(z - \mu)^2)$

Example: General Normal Random Variable



Unconditional Probabilistic Models

- Definition: A parametric probability model $\mathbb{P}$ for a continuous random variable Z with with parametric probability density function p and parameter space Φ is the set of all probability density functions generated by p and Φ:

$$\mathbb{P} = \{p_\phi(Z = z) | \phi \in \Phi\}$$

Parameter Estimation

Let $\mathcal{L}(\mathcal{D}, \phi)$ be a loss function that measures how “close" a data set $\mathcal{D}$ is to a parametric probability density function with parameters ϕ. Given such a loss function, we can learn the parameters as follows.

The Parameter Estimation Problem

$$\hat{\phi} = \arg\min_{\phi \in \Phi} \mathcal{L}(\mathcal{D}, \phi)$$

This selects the best distribution expressible by the model $\mathbb{P}$ according to the loss function *L*.

Maximum Likelihood Estimation

- The most common such loss function used to estimate the parameters of probabilistic models is the negative log likelihood function:

$$nll(\mathcal{D}, \phi) = -\sum_{n=1}^{N} \log p_\phi(Z = z_n)$$

Maximum Likelihood Estimation

- This loss function derives from the idea of selecting the parameters ϕ that assign the highest probability to the data.

$$\hat{\phi} = \underset{\phi \in \Phi}{\arg\max}\ p_\phi(\mathcal{D})$$

$$= \underset{\phi \in \Phi}{\arg\max}\ \log p_\phi(\mathcal{D})$$

$$= \underset{\phi \in \Phi}{\arg\max}\ \log \left(\prod_{n=1}^{N} p_\phi(Z = z_n) \right)$$

$$\underset{\phi \in \Phi}{\arg\max}\ \sum_{n=1}^{N} \log p_\phi(Z = z_n)$$

$$= \underset{\phi \in \Phi}{\arg\min}\ - \sum_{n=1}^{N} \log p_\phi(Z = z_n)$$

Maximum Likelihood Estimation

- The process of selecting model parameters to minimize the negative log likelihood (or equivalently maximize the likelihood) of observed data is referred to as *maximum likelihood estimation*.
- The idea that we should select parameters to minimize the negative log likelihood of the observed data is referred to as the *likelihood principle*.
- The resulting learned (or estimated or fit) parameters are referred to as the *maximum likelihood estimate* or MLE.
- The use of maximum likelihood for probabilistic model estimation was developed by R.A. Fisher in the 1920s.

Example: Normal Mean

- Let $p_\phi(Z = z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(z-\mu)^2\right)$
- Suppose we have a data set $\mathcal{D} = [z_1, \ldots, z_N]$ and we want to find the MLE of μ.

- The negative log likelihood function is:

$$nll(\mathcal{D}, \mu, \sigma) = \sum_{n=1}^{N}\left(\frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}(z_n - \mu)^2\right)$$

- The MLE is $\hat{\mu} = \frac{1}{N}\sum_{n=1}^{N} z_n$.

Probabilistic Regression

- In probabilistic regression, our goal is to model the true probability distribution of a regression target $y \in \mathcal{Y}$ given the inputs $\mathbf{x} \in \mathcal{X}$.
- If $\mathcal{Y}$ is continuous, our goal is to model $p_*(Y = y|\mathbf{X} = \mathbf{x})$ with a conditional parametric probability density function $p_\theta(Y = y|\mathbf{X} = \mathbf{x})$.

Generating Probabilistic Regression Models

- We can very flexibly generate probabilistic regression models by combining unconditional probability models with models that predict their parameter values.
- We can select different probability models to provide different distributions over the output space $\mathcal{Y}$.
- We can use any type of model to predict the unconditional probability model parameters including both linear and non-linear models.
- The parameter prediction model may need to include a transformation to ensure that the predicted parameter values always fall in the parameter space of the selected unconditional probabilistic model.

Learning Probabilistic Regression Models

- So long as all model components are differentiable functions, we can learn the model parameters θ by minimizing the conditional negative log likelihood function given a data set $\mathcal{D}$:

$$nll(\mathcal{D}, \theta) = -\sum_{n=1}^{N} \log p_\theta(Y = y | \mathbf{X} = \mathbf{x})$$

- This is technically maximum conditional likelihood estimation, but is also often just referred to as maximum likelihood estimation.

Example: Probabilistic Linear Regression

- Suppose that $\mathcal{Y} = \mathbb{R}$ and $\mathcal{X} \in \mathbb{R}^D$.
- Unconditional Model: $p_\phi(Y = y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$
- Conditional Model:

$$p_\theta(Y = y|\mathbf{X} = \mathbf{x}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y - f_w(\mathbf{x}))^2\right)$$

- Parameter Prediction Function: $f_w(\mathbf{x}) = \mathbf{x}\mathbf{w}$
- Model Parameters: $\theta = [\mathbf{w}, \sigma]$
- Negative Log Likelihood:

$$nll(\mathcal{D}, \theta) = -\sum_{n=1}^{N} \log p_\theta(Y = y_n|\mathbf{X} = \mathbf{x}_n)$$

Example: Probabilistic Linear Regression

$$
\begin{aligned}
nll(\mathcal{D}, \theta) &= -\sum_{n=1}^{N} \log p_\theta(Y = y_n | \mathbf{X} = \mathbf{x}_n) \\
&= -\sum_{n=1}^{N} \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{1}{2\sigma^2} (y_n - f_w(\mathbf{x}_n))^2 \right) \right) \\
&= -\sum_{n=1}^{N} \left(-\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y_n - \mathbf{x}_n \mathbf{w})^2 \right) \\
&= \sum_{n=1}^{N} \left(\frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} (y_n - \mathbf{x}_n \mathbf{w})^2 \right)
\end{aligned}
$$

This model has a closed form solution for the optimal $\mathbf{w}$ (identical to the OLS solution), as well as a closed form solution for the optimal σ with $\mathbf{w}$ fixed.

Example: Probabilistic Non-linear Regression

- Suppose that $\mathcal{Y} = \mathbb{R}$ and $\mathcal{X} \in \mathbb{R}^D$.
- Unconditional Model: $p_\phi(Y = y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$
- Conditional Model:

$$p_\theta(Y = y|\mathbf{X} = \mathbf{x}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y - f_{\mathbf{w}}(\mathbf{x}))^2\right)$$

- Parameter Prediction Function: $f(\mathbf{x})$ = an MLP, CNN, RNN, transformer, etc.
- Negative Log Likelihood:

$$nll(\mathcal{D}, \theta) = -\sum_{n=1}^{N} \log p_\theta(Y = y_n|\mathbf{X} = \mathbf{x}_n)$$

Example: Heteroskedastic Regression

- Suppose that $\mathcal{Y} = \mathbb{R}$ and $\mathcal{X} \in \mathbb{R}^D$.
- Unconditional Model: $p_\phi(Y = y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y - \mu)^2\right)$
- Conditional Model:

$$p_\theta(Y = y|\mathbf{X} = \mathbf{x}) = \frac{1}{\sqrt{2\pi g_{\mathbf{v}}(\mathbf{x})^2}} \exp\left(-\frac{1}{2g_{\mathbf{v}}(\mathbf{x})^2}(y - f_w(\mathbf{x}))^2\right)$$

- Parameter Prediction Functions: $f_w(\mathbf{x}) = \mathbf{x}\mathbf{w}$, $g_{\mathbf{v}}(\mathbf{x}) = \log(1 + \exp(\mathbf{x}\mathbf{v}))$
- Model Parameters: $\theta = [\mathbf{w}; \mathbf{v}]$
- Negative Log Likelihood:

$$nll(\mathcal{D}, \theta) = -\sum_{n=1}^{N} \log p_\theta(Y = y_n|\mathbf{X} = \mathbf{x}_n)$$

Making Predictions

- Given a probabilistic regression model, we can produce an estimate of the conditional probability of y given $\mathbf{x}$ by plugging the estimated parameters $\hat{\theta}$ into the model.
- In the case of continuous y, we can predict different functions of the conditional distribution. The most commonly used prediction is the conditional mean of y:

$$\hat{y} = E_{p(Y=y|\mathbf{X}=\mathbf{x},\hat{\theta})}[y] = \int_{\mathcal{Y}} y p(Y = y|\mathbf{X} = \mathbf{x}, \hat{\theta}) dy$$

- For regression models based on the normal distribution, we have:

$$\hat{y} = E_{p(Y=y|\mathbf{X}=\mathbf{x},\hat{\theta})}[y] = f_{\mathbf{w}}(\mathbf{x})$$